

FREE

SSRI Withdrawal Is Mitochondrial Dysfunction

Installment seven in our series on understanding the truth about SSRIs.



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In analyzing the biochemical data of clients over the last year I have worked on two cases involving catastrophic failures of energy metabolism that occurred in the wake of SSRI withdrawal. In both cases, *switching* SSRIs from one to another also seemed to play a role in precipitating energetic dysfunction.

Here we will examine the concept that both SSRI *withdrawal* and SSRI *switching* can cause mitochondrial dysfunction.

This is the seventh installment in our [series](#) on the truth about serotonin and SSRIs and builds on the information in the last installment, [SSRIs Are Mitochondrial Drugs](#).

This is educational in nature and not medical or dietetic advice. See [terms](#) for additional and more complete disclaimers.

The first individual had no issues being on Paxil long-term nor going off Paxil for the first time, but when he went on Paxil a second time he developed tachycardia and muscle cramps and spasms during sleep. Switching to Lexapro made these worse. *Withdrawing* from Lexapro then added to these problems a whole list of new problems: sexual dysfunction, peripheral neuropathy, shortness of breath, anxiety, muscle cramping, numbness, and excessive urination.

The second individual developed sexual dysfunction on Zoloft and chronic fatigue in the wake of Zoloft withdrawal in his 20s. At its worst point, he spent a year not being able to stand in the shower and a month needing to be bathed by another person and needing a portable bedside toilet.

Despite these problems onsetting during the *withdrawal* from Zoloft, Prozac immediately caused extreme spaciness and a low dose of Celexa caused restlessness. He is not confident that Zoloft would not have caused the same problems at that time because he felt hypersensitive to everything. However, the hypothesis that switching SSRIs did indeed worsen his situation should be considered, because SSRIs have different impacts on mitochondrial function.

Both people received individualized protocols based not on generalized information about SSRI withdrawal, but based on mitochondrial testing, whole genome sequencing, and comprehensive biochemical data.

The second just received his protocol so has no results to report yet.

The first received his protocol in November of last year. Despite “having gone to see dozens of specialists who could not tell me what was going on,” six weeks into his

protocol he reported “the first time I felt normal in over two years.”

At the six-month mark, he wrote “six months later, I have regained quality of life and am able to do everything I used to be able to.”

After 8.5 months, he wrote “I've still been pretty amazed how effective your protocol has been. On the whole I feel like a whole different person from a year ago. It's wild. I've also gained close to 15 pounds in lean muscle mass since January. Making gains in the gym daily and I think I can end the year with another 10 pounds of mass. Been crushing it at work and also traveling and feel comfortable in my body for the first time in 2-3 years. Excited to see what continued progress I can make!”

The [Mitome](#) results of the two individuals were very different. Importantly, neither of them had Mitome tested in the immediate wake of SSRI withdrawal, so in each case the results reflect the combination of genetics, nutritional status, and prior experience, including the catastrophic effects of SSRIs and being off them for several years. The first showed a pattern associated with deficient methylation expected to *suppress* the hypoxia response, while the second showed a pattern associated with the impact of an *excessive* hypoxia response.

[Whole genome sequencing](#) suggested the first individual's results were caused by a heterozygous defect in SLC25A26, which prevents methyl groups from being transported into the mitochondria, while the second individual's results were caused by a heterozygous gain-of-function polymorphism in a receptor for TNF-alpha that creates a chronically increased inflammatory tone, which primes cells to easily activate a hypoxia response that suppresses mitochondrial respiration.

The second case is rather straightforward to explain: the hypoxia response is antagonized by sertraline (Zoloft) but exacerbated by fluoxetine (Prozac) and citalopram (Celexa). Thus, while the second individual remains deeply uncertain whether these different SSRIs had genuinely different impacts on him at the time they were tried, the response is consistent with the experimental evidence suggesting these different SSRIs have disparate impacts on the hypoxia response.

The first case is less straightforward to explain because paroxetine (Paxil) appeared to be harmless upon first use and withdrawal but the negative reaction to the second use implies that *yo-yoing* on paroxetine can be particularly harmful. This yo-yoing effect could be seen as a disorganized response to the conflicting positive and negative effects of paroxetine on mitochondrial function:

- Paroxetine has the least activation of the sigma-1 receptor out of any SSRI, which suggests its ability to increase extracellular serotonin will lead to a high ratio of mitochondrial biogenesis to mitophagy, and this is consistent with rodent studies showing it increases markers of mitochondrial density.
- While the hypoxia response usually opposes mitochondrial biogenesis, the unnatural level of extracellular serotonin achieved with SSRIs will activate both processes. Paroxetine's lack of sigma-1 activation makes it deplete brain serotonin instead of increasing brain serotonin like the SSRIs that are powerful sigma-1 activators, which means it will increase the anti-respiration aspects of the hypoxia

response without supporting the pro-respiration aspects mediated by intracellular serotonin and melatonin.

- This individual was probably hyper-tolerant of the hypoxia response promotion due to his idiosyncratic mitochondrial pattern expected to suppress that response. So, the initial experience of paroxetine may have been simply balancing out the hypoxia response in his particular case, while its promotion of mitochondrial biogenesis may have been quite useful.
- However, withdrawing from paroxetine could have caused a rebound *decrease* in mitochondrial density that created a greater vulnerability to the toxic actions of paroxetine that became evident upon reusing the drug. This could include the imbalanced hypoxia response, direct inhibition of mitochondrial serotonin receptors, and direct inhibition of respiratory chain complexes. While he may have initially tolerated the imbalanced hypoxia response, this tolerance could have been lost as a result of the rebound decrease in mitochondrial biogenesis.

Lexapro (escitalopram) shares with paroxetine (Paxil) the increase in extracellular serotonin and the promotion of respiration-impairing aspects of the hypoxia response. However, it is a ten times more powerful sigma-1 activator. It is nowhere near as powerful as fluvoxamine or sertraline, but it is much more powerful than paroxetine. This would be expected to balance mitochondrial biogenesis with mitophagy and to balance the hypoxia response with the preservation of respiration caused by intracellular serotonin and melatonin. However, it is also the most mitochondrially toxic of the SSRIs when compared head-to-head using isolated mitochondria. It is especially toxic toward complex IV compared to other SSRIs, and his complex IV was already hurt by his impairment in methyl group transport.

If the paroxetine yo-yoing effect was indeed caused by a rebound drop in mitochondrial density, then escitalopram's stimulation of mitophagy via the sigma-1 receptor may have aggravated this with a further drop in mitochondrial density.

One thing we have to understand here is that, while the *regulation* of mitochondrial biogenesis is ordinarily stimulated by an energetic deficit, which signals a need for more mitochondria, the actual *process* of building new mitochondria is incredibly energy-intensive. It is anabolic. It cannot occur without energy no matter what the regulation says should be happening.

Given all that, this is the likely sequence of events:

- The yo-yoing of paroxetine caused a rebound drop in mitochondrial density after the first withdrawal that flipped the response to a second use in favor of vulnerability to its mitochondrial toxicity.
- This then caused energetic dysfunction that prevented the second use of paroxetine from rebuilding the lost mitochondria.
- This then allowed the relative balance favoring of mitophagy by escitalopram to worsen this.
- The lower mitochondrial content of tissues primed them to respond poorly to the higher mitochondrial toxicity of escitalopram.

The astounding benefits he obtained from his protocol were NOT based on trying to fix the catastrophic destruction induced by SSRI use, withdrawal, and switching, but rather on a coherent theory of his biochemical idiosyncrasies at the time I worked on his case informed by insights into which were genetic in nature based on the analysis of his whole genome sequencing. The lion's share of the benefits were deducible from [Mitome](#) alone, though [whole genome sequencing](#) and [nutritional/biochemical](#) analysis enhanced the interpretation.

How much of this is generalizable across SSRI withdrawal? Let's take a look at the literature.

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SSRI Discontinuation Syndrome

[SSRI discontinuation syndrome](#) occurs in over half of people who stop SSRIs according to observational studies, and in about twenty percent of people in randomized controlled trials. Even using the smaller number from the trials, there are millions of people worldwide who are undergoing cessation of SSRIs at any given moment, suggesting there are always hundreds of thousands of people experiencing SSRI discontinuation syndrome in any given moment.

This can involve irritability, anxiety, mood problems, crying, dread, suicidal ideation, insomnia, nightmares, excessive dreaming, lethargy, fatigue, headache, tremor, sweating, anorexia, flu-like symptoms, nausea, vomiting, diarrhea, pain, numbness, tingling, feeling like something is crawling on the skin, electric shocks running through the brain or body, rushing noises, visual traces (seeing something persist when it is no longer there, or seeing moving objects leaving illusory streaks of light behind them, etc), dizziness, light-headedness, vertigo, confusion, difficulty concentrating, amnesia, genital hypersensitivity, and premature ejaculation.

The sexual side effects of SSRI discontinuation are not considered to include the genital numbness and erectile dysfunction characteristic of post-SSRI sexual dysfunction (PSSD), but as we will consider below PSSD can indeed become a long-

lasting SSRI side effect that doesn't start until after SSRIs are stopped, meaning that a subset of PSSD is indeed a subset of SSRI discontinuation syndrome.

SSRI discontinuation syndrome tends to require more than one week and perhaps six to eight weeks of SSRI treatment as a pre-requisite. It occurs more often in men than women, younger individuals than older, in those that stop SSRIs abruptly instead of tapering, and more often with Paxil (paroxetine) than Prozac (fluoxetine) or [Lexapro \(escitalopram\)](#). It can appear as early as 24 hours after the last dose, usually resolves in two to three weeks, but can last longer, even months or years.

[Analysis of internet forums](#) shows that “protracted withdrawal syndrome” can last anywhere from five months to almost fourteen years. The average person reporting such protraction of their discontinuation symptoms was using SSRIs for eight years prior to withdrawal.

[A survey](#) of people in the UK-based National Health Services found that whether withdrawal problems occurred increased progressively as the length on the SSRI went up: 64% for those using them less than six months; 86% for those using them from 6 months to two years; and 96% of those who had used them for more than two years.

38% reported trying to stop but being unable to do so.

Those reporting “severe” withdrawal symptoms were much more likely to have used SSRIs for a longer period of time: 7% of those using them less than six months; 15% of those using them between six months and two years; and 25% of those using them for longer than two years.

Withdrawal symptoms lasting longer than a year was reported by 7% of those who had been on SSRIs less than six months and 12% of those who had been on them for more than two years.

The most common “non-emotional” problems reported by the majority were headache, derealization, depersonalization, and lightheadedness. 28% reported “brain zaps,” 27% reported muscle cramps, 27% reported nausea or vomiting, 24% reported diarrhea, and 30% reported trouble walking stably.

An [in-depth study](#) of “brain zaps” where people were approached through forums on mentalhealthdaily.com found that the onset to the first zap was eight days with fluoxetine and two days with paroxetine, suggesting the big difference in half-life of these two drugs primarily led to a difference in how long it took to experience the zaps.

52% said the “zaps” felt like an electric shock inside the head, 18% said it felt like a vibration, 12% said they could hear the zaps, 4% said they could see the zaps, 7% said it felt like a pop or a twitch, and a small number of people described it with such terms as a strobe without a light, a crackle, their brain skipping a beat or blinking, or as if the video of their mind was buffering.

The most common triggers of brain zaps were moving the eyes or the head from side to side.

The most up-to-date reviews mention NOTHING about mitochondria in their explanations of this syndrome. The common explanation for why Paxil has a higher prevalence of discontinuation syndrome is that it is much more rapidly cleared from the body. This is an extremely unsatisfactory explanation, however, because this should be easily solved with a more drawn-out tapering.

These reviews mention NOTHING about the sigma-1 receptor. In fact, the most [recent review](#) on the mechanisms of SSRI discontinuation syndrome is part of an anthology titled “Emerging Neurobiology of Antidepressant Treatments” and is written by one of the editors of the anthology yet the entire anthology is over 300 pages and does not mention sigma-1 receptors even once. Thus, there is no discussion of how Paxil stands out as the weakest activator of the sigma-1 receptor out of all SSRIs, ten times weaker than Lexapro and twenty times weaker than Prozac. This positions it as uniquely powerful in its ability to *deplete* total brain serotonin, making the sudden absence of its ability to sustain extracellular brain serotonin much harder to deal with.

[In rats](#), fluoxetine produces an initial decrease in brain serotonin lasting one to three days, which then recovers to a slightly higher than baseline level thereafter. Withdrawing from fluoxetine causes a much larger initial drop in brain serotonin that takes seven days to recover from.

This is consistent with fluoxetine being a moderate sigma-1 activator. Initially, a higher extracellular serotonin concentration causes greater activation of serotonin receptors that exert negative feedback on serotonin synthesis. However, its activation of the sigma-1 receptor then causes a broad increase in intracellular serotonin synthesis. The rats then become dependent on the constant sigma-1 activation to counter the constant overactivation of serotonin receptors. When fluoxetine is withdrawn, the drop in sigma-1 activation causes a sudden drop in serotonin synthesis, but then the return to normal extracellular serotonin signaling allows this to even out.

By contrast, treatment of mice for twelve days with [paroxetine](#) causes a decrease in brain serotonin and withdrawal leads to a rebound above baseline at least as early as day 2. In some areas of the brain, the rebound is mild and increases further into day 5. In others, it is pronounced, and it levels off by day 5.

The authors of these studies use individual SSRIs as a representative of the drug class and then generalize about the discontinuation syndrome. This leads to fundamentally incoherent conclusions.

For example, the authors of the latter study — done by the same group that wrote the first-cited review, one of whom was an editor of the anthology — suggest that SSRI discontinuation syndrome is driven by rebound overactivation of the serotonin system.

Separately, they claim that paroxetine has a higher rate of discontinuation syndrome than fluoxetine because it leaves the body faster.

If the latter claim is true, paroxetine should do the same thing to brain serotonin during withdrawal as fluoxetine, but should do it faster. Yet they simply have opposite

effects: equilibration to fluoxetine leads to a slightly higher brain serotonin level than baseline, and equilibration to paroxetine leads to a considerably lower level than baseline; discontinuation of fluoxetine leads to a big drop in brain serotonin and discontinuation of paroxetine leads to an increase.

This recent paper, like the more than 300-page anthology, makes no mention of sigma-1 receptors. Yet these receptors easily explain the results: fluoxetine maintains serotonin synthesis by activating this receptor and paroxetine fails to do so. Therefore, the negative effect on brain serotonin of activating the extracellular serotonin receptor dominates the effect of paroxetine, so serotonin goes down on the drug and up during withdrawal; the sigma-1 activation caused by fluoxetine is the slightly dominant effect, so serotonin goes slightly up on the drug and down during withdrawal.

Yet they are both associated with SSRI discontinuation syndrome, making rebound increases or decreases in serotonin poor explanations of the syndrome, at least in isolation and treating the syndrome as if it is a single pathological entity.

Another hypothesis to explain the higher rate of discontinuation syndrome with Paxil is that it has anticholinergic activity by binding to and inhibiting acetylcholine receptors, so its withdrawal will cause a rebound excess of acetylcholine activity. But this is a unique function of Paxil among the SSRIs and is similar to the older tricyclic antidepressants, whereas between 20% and a majority of all SSRI users get this syndrome, and the symptoms expected from a cholinergic rebound are limited to a relatively small portion of the discontinuation symptoms. So it could play a minor role in Paxil withdrawal but not an overall major role in the syndrome.

Paxil also has modest inhibition of dopamine and norepinephrine reuptake, so Paxil withdrawal could lower activity of these neurotransmitters, but the criticisms of the anticholinergic hypothesis also apply here.

The most up-to-date reviews admit this syndrome is poorly understood, and recommend tapering SSRIs, but also acknowledge that “present guidelines for managing SSRI discontinuation are based on dose tapering and not a knowledge of the neural mechanisms underlying discontinuation.”

In making this statement, they also betray the fact that they are intellectually trapped into the belief that the mechanism is fundamentally “neural” when the answer lies in accepting that SSRIs are whole-body drugs acting primarily on cells that aren’t neurons, and that what they do to neurons and what they do to other cells fundamentally involves messing with serotonin and melatonin’s impact on mitochondria as well as the SSRIs themselves entering the cells and activating other receptors such as the sigma-1 receptor.

Post-SSRI Sexual Dysfunction

Sexual dysfunction was at first recognized as occurring on SSRIs. [Post-SSRI Sexual Dysfunction \(PSSD\)](#) was recognized after evidence accumulated that this sexual dysfunction could persist after discontinuing SSRIs. The most commonly reported problems in men are genital anesthesia (total loss of any feeling in the genitals), loss of

sexual pleasure, weak orgasms, decreased libido, erectile dysfunction, and premature ejaculation.

PSSD is often associated with general anhedonia, apathy, and poor mood.

Genital anesthesia, and, in women, loss of sexual sensitivity in the nipples, are according to some reviews specific problems associated with PSSD that are not reported in other forms of sexual dysfunction. However, genital anesthesia also accompanies similar syndromes that endure after the use of [finasteride](#), which inhibits the conversion of testosterone to the more potent DHT, and [isotretinoin](#), a compound that mimics some aspects of vitamin A but otherwise acts as a vitamin A antagonist. The [incidence of PSSD in SSRI users](#) has been estimated to be 0.46%.

Researchers and clinicians dealing with PSSD have speculated that genital anesthesia results from damaged spinal cord neurotransmission; that decreased libido results from impaired dopamine transmission in the brain; that erectile dysfunction could result from SSRIs altering the function of oxytocin or androgens; and that weak orgasms result from decreased oxytocin and beta-endorphin release. They have further speculated that multiple rather than single uses of SSRIs, preexisting disorders of glutamate neurotransmission, MTHFR variants, and abuse of recreational drugs could all increase the risk of PSSD.

PSSD is [consistently described](#) in most reviews not as sexual dysfunction arising after the withdrawal of SSRIs, but rather sexual dysfunction arising often within days or weeks of starting SSRIs, occasionally within 30 minutes of the first dose, and either persisting or worsening after withdrawal of the medication. However, [more broad-minded reviews](#) see a more general syndrome associated mainly with use of SSRIs, isotretinoin, and finasteride, and possibly with non-SSRI neuropsychiatric drugs like buprenorphine, ondansetron, quetiapine, lithium, and haloperidol, and note that it can have its onset after the cessation of the drug. Most of the non-SSRI neuropsychiatric drugs had been co-prescribed with SSRIs, however, so the best high-level perspective is that this is a syndrome that can have its onset soon after the initial use, any time during the use, or after the cessation of SSRIs, finasteride, or isotretinoin. The syndrome can last for as little as weeks or months, but has been documented to last, in the longest case, for at least 18 years.

Even very recent reviews have a nearly complete dearth of useful actionable information or genuine insights into the causation of these syndromes. Speculations about the mechanisms are at a level of proximate cause (for example, decreased neurotransmission through the spinal cord driving genital anesthesia), not ultimate cause, and therefore shed little light on what could be done about the problems. Reviews claim that most case reports fail to support any benefit of testosterone replacement therapy, PDE5 inhibitors, dopamine-boosting drugs, or any other drugs that have been tried.

There is, by contrast, [support for the use of](#) bupropion and PDE5 inhibitors from subjects with SSRI-induced sexual dysfunction who are still on SSRIs. This includes Viagra improving erectile dysfunction in a meta-analysis of multiple trials; a single, small, double-blind study using bupropion as an adjunct therapy for broader sexual dysfunction; and a small number of case reports using bupropion similarly. The

[bupropion trial](#) selected only individuals who were presently using SSRIs and had been using them for at least three months. There was a 20% increase in the desire/frequency subscore of the sexual function test, but there was no statistically significant difference in the total score. Notably the baseline scores were 40 out of 70, where the cutoff for sexual dysfunction is 47 for males and 41 for females, indicating only mild sexual dysfunction.

The commonality of this syndrome across exposure to SSRIs, finasteride, and isotretinoin suggests these drugs could share a primary mechanism in common. Given the importance of testosterone to sexual function, one possibility is that they all do what finasteride is designed to do: lower the conversion of testosterone to DHT.

In [one study](#), isotretinoin in the amount of 1 milligram per kilogram of bodyweight for 20 weeks lowered DHT in women but not men; in a [second study](#), 0.75 mg/kg isotretinoin for twelve weeks lowered DHT by 21% in men, and the resulting [ratios of metabolites](#) suggested it inhibits 5-alpha-reductase just like finasteride does. However, in vitro, it has been shown to inhibit DHT production through an alternative pathway dependent on [3-alpha-hydroxysteroid dehydrogenase](#) and no in vitro studies have examined whether it directly inhibits 5-alpha-reductase or decreases its expression.

Bizarrely, despite the existence of this syndrome and its establishment in the literature, there are no studies examining the effect of SSRIs on DHT.

Viagra prevents the breakdown of cGMP, a mediator of erections. cGMP synthesis is initiated by nitric oxide. Nitric oxide is also the mediatory of the parasympathetic branch of the sexual nervous system which mediates the totality of arousal. Several serotonin receptors are capable of increasing or decreasing nitric oxide synthesis. Sertraline and escitalopram have both been shown to decrease nitric oxide production. Therefore, SSRIs probably decrease nitric oxide production on average, and Viagra helps to counteract the effects of this on cGMP.

Bupropion has complex actions but one is to increase the activity of dopamine and norepinephrine by inhibiting their reuptake from synapses, so this can easily explain its benefit to sexual function across the board.

However, the fact that these drugs help does not necessarily tell us anything about the root cause of PSSD. Viagra will cause erections in any man, and bupropion [improves sexual function](#) even in women who that are not depressed and have never used SSRIs.

Prozac Withdrawal Causes Mitochondrial Dysfunction in Mice

The only study I am aware of that looked at the impact of SSRI withdrawal on mitochondrial function was a study administering Prozac to rats. This one study appears to have generated several papers, including one on [respiratory chain complexes](#) and another on [citrate synthase](#), which is a citric acid cycle enzyme that can serve as a useful but imperfect marker of mitochondrial density.

They administered a low and high dose for 28 days.

Two hours after the *first* dose, the Prozac increased complex I in the hippocampus. **On day one, Prozac was beneficial.**

Two hours after the *last* dose, 28 days later, Prozac had no effect at all. **After a month, the benefit had worn off.**

Twenty-four hours after the last dose, however, the rats withdrawing from the high dose of Prozac had reduced complex II + III activity in the striatum and those withdrawing from both doses had reduced complex IV activity in the hippocampus. **During withdrawal, the respiratory chain was harmed.**

On day one, Prozac increased citrate synthase in the striatum. In the hippocampus and prefrontal cortex it seemed greater than in controls but it wasn't statistically significant. Two hours after the last dose citrate synthase seemed lower in the striatum, but it wasn't statistically significant. 24 hours after the last dose, citrate synthase appeared unchanged across the board. **Overall there seemed to be a mild stimulation of mitochondrial biogenesis that wore off over time.**

The net result of this is that Prozac caused an acute improvement in mitochondrial function. Over time, the rats became dependent on it. They adjusted to the drug, and then after 24 hours of withdrawal they had new-onset mitochondrial dysfunction.

Near Infrared Light Reverses Genital Anesthesia

Another hint that SSRI withdrawal is mediated by mitochondrial dysfunction is a case report suggesting genital anesthesia can be reversed with near infrared light.

[One case report](#) showed that low-powered laser in the near infrared range on the scrotum and spine led to a 40% increase in penile sensitivity.

Near infrared is obtainable from the sun as well as home devices like near infrared saunas or lamps. [Infrared light](#) acts in part by feeding photons to complex IV (cytochrome oxidase), which improves its ability to facilitate ATP production. Further, near infrared also [structures mitochondrial water](#) to improve its viscosity in a way that facilitates greater production of ATP by ATP synthase.

Mitochondrial Dysfunction Tanks Testosterone

[In male rats](#), inhibiting complex I with the fish poison rotenone depletes testosterone.

According to Chapter 10 of Saudubray, *Inborn Metabolic Diseases: Diagnosis and Treatment*, mitochondrial disorders can result in infertility and hypogonadism.

This can be explained through several mechanisms.

First, the synthesis of testosterone is anabolic and therefore requires energy input. If you tank your energy production, you do not have energy to make testosterone.

Second, testosterone is itself anabolic. If you tank your energy production, you will tell your hypothalamus that you cannot *afford* to make testosterone, because if you do, you will increase your synthesis of muscle, which is extremely energetically expensive.

Worse, you could wind up having sex.

For men, you could get someone pregnant, and then be responsible for spending precious energy providing for the anabolism of that new human and its future upbringing.

For women, you get put on the hook for spending your energy on making an entirely new human and then feeding it milk for one to three years thereafter.

Therefore, energetic deficiency makes it more difficult to make testosterone and makes the brain or sex organs restrain its production.

Mitochondrial Dysfunction Causes Neurological Dysfunction

The brain is only two percent of our bodyweight but consumes twenty percent of its oxygen.

Energy must be invested in the anabolic synthesis of myelin, neurotransmitters, and other brain materials and to maintain neurotransmitter function from each fraction of a second to the next.

The release of a neurotransmitter into a synapse, its action on the downstream neuron, its clearance from the synapse, the transmission of an action potential down the neuron, and the trigger for another neurotransmitter to be released at the other end all involves the exquisite control of a symphony of calcium, magnesium, sodium, chloride, and potassium ions that are pumped to one side of a membrane and unleashed to flow back to the other side over and over again, where all the pumps are fueled by the energy contained within the ATP made by the neuronal mitochondria.

According to Chapter 10 of Saudubray, *Inborn Metabolic Diseases: Diagnosis and Treatment*, mitochondrial disorders can result in the following problems reported during SSRI withdrawal: ataxia, cramps, and peripheral neuropathy.

Muscle contraction occurs when sodium- and potassium-based neural signals cause calcium to be released at the end of a neuron, which causes acetylcholine to be released from the neuron onto the muscle, which then causes calcium to be released inside the muscle cell, which then signals the contraction. Muscle relaxation occurs when ATP puts the calcium back where it belongs. Magnesium is needed to support ATP, explaining the role of magnesium in muscle relaxation. If any of these ions are deficient or are not pumped where they are supposed to go using energy from ATP, you get cramps.

Ataxia can occur from energetic failure in various parts of the nervous system or inner ear responsible for balance and coordination.

The causation of peripheral neuropathy has been reviewed [here](#) and [here](#). It is a form of “ectopic signaling,” meaning signaling that is occurring in the wrong place. You can have a physical injury where sprouts of regenerating sensory receptors fire abnormally because they have lost normal connections, or because demyelinated nerves can bind together abnormally and fire abnormally, or can allow their signals to be discharged

horizontally, propagating among a broader set of nerves than would normally be impacted.

Physical injury can be seen as a forceful undoing of the work of mitochondrial energy production. This energy is invested in creating order, and when its work is forcefully undone, relative chaos ensues.

If you have primary energetic failure, you get the same result: rather than a physical injury disrupting the work of mitochondrial energy production, the work cannot be done in the first place. Thus, disordered ion flows and neural connections produce ectopic signaling.

The specific abnormal sensations generated during ectopic signaling takes on flavors from the subconscious, from fears, or from the history of past sensations, which is why one person can experience pain and another tingling and another electric shocks.

Some of the problems of SSRI withdrawal are more reminiscent of psychedelics than the common symptoms of mitochondrial dysfunction. For example, visual traces, being able to see or hear brain zaps, and hearing whooshing sounds that aren't there.

While electric shock-like pain is found in many disorders and can be seen as similar in its causation to peripheral neuropathy, "brain zaps" themselves appear to be a phenomenon relatively isolated to SSRI withdrawal.

Psychedelics act on 5-HT_{2A} receptors to induce psychedelic experiences, but serotonin does not normally induce these experiences even though it activates these receptors. [Research suggests](#) that this is because psychedelics but not serotonin *indiscriminately enter the cell* and bind to *intracellular* 5-HT_{2A} receptors, located at least partly on the Golgi apparatus, and that they can stay activating these receptors for prolonged periods of time. By contrast, the areas of the brain involved in these perceptual modifications do not substantially express serotonin transporters outside those involved in reuptake of synaptic serotonin under normal conditions. Therefore, serotonin doesn't usually enter these cells. However, if mice are engineered to highly express serotonin transporters in the relevant neurons the serotonin can enter the cell and it can induce these types of perceptual changes.

As covered in [SSRIs are Mitochondrial Drugs](#), activation of the sigma-1 receptor increases expression of serotonin transporters in the cell membrane. These are normally activated under whole-body or cellular stress of multiple types. The natural ligands are unknown but may include DMT (the chemical from ayahuasca that we also make in small amounts endogenously), choline, sphingolipids, myristic acid, and DHEA-sulfate.

It is likely that in hypoxic stress, monamine oxidase activity declines and DMT rises, which would allow it to activate sigma-1 receptors. However, these conditions also facilitate the movement of serotonin into the mitochondria where it is converted to melatonin rather than acting as a persistent activator of Golgi 5-HT_{2A} receptors. Further, the response is coordinated to normalize energy production under hypoxia as quickly as possible. Nevertheless, hypoxia probably *sometimes* activates these receptors to cause perceptual abnormalities, since some people can develop psychosis from

[altitude](#). Psychological stress, which activates sigma-1 receptors, is also a [major contributor](#) to psychotic breaks.

The question arises what on earth serotonin receptors are doing inside the cells of neurons that do not usually express serotonin transporters. Perhaps they are sitting there waiting for LSD or mushrooms to arrive. More likely, various forms of stress activate serotonin entry into the cell, where it does activate those receptors but not usually in a sustained and isolated way that facilitates a psychedelic trip.

SSRI withdrawal could induce sustained activation of these receptors in a few ways we can brainstorm:

- Chronic activation of the sigma-1 receptor may cause a long-lasting increase in serotonin transporters on the membranes of these neurons.
- Chronic deprivation of intracellular serotonin in and of itself causes a long-term feedback loop increasing the serotonin transporter expression on these neurons, primarily to secure sufficient mitochondrial melatonin.
- Mitochondrial dysfunction in the wake of serotonin withdrawal leads to sigma-1 activation, increasing the serotonin transporters on the membranes of these neurons.

The relative specificity of this to SSRI withdrawal could simply result from a bias toward having greater impacts on certain parts of the brain with high concentrations of hypoxia response-promoting serotonin receptors.

The “brain zaps” being associated with side-to-side head and eye movements was suggested by the authors of that study to be related to activation of the vestibulo-ocular reflex, whose job is to stabilize vision by making eye movements compensate for head movements.

This reflex is [powerfully activated](#) by serotonin, Prozac, and the “God molecule” psychedelic from toad venom, 5-methoxy-DMT. The commonality of these suggests it is mediated by activation of extracellular serotonin receptors.

Brain zaps are probably a form of “ectopic signaling” that is very similar in nature to peripheral neuropathy, but is a result of specific disruptions to ion gradient maintenance in the neural circuitry of the vestibulo-ocular reflex. Thus, the reflex activates with a lateral eye or head movement and then ions spill into the wrong places and initiate a perceptual abnormality.

This likely reflects mitochondrial dysfunction that has essentially gravitated to an area of the brain where extracellular serotonin is particularly important due to the long-standing use of a mitochondrial drug that is centrally involved in increasing extracellular serotonin.

How Common Is Mitochondrial Dysfunction as a Mediator of SSRI Discontinuation Syndrome and PSSD?

While we cannot answer this with rigorous research, this is because serotonin is almost universally perceived as a “neural” and “psychiatric” molecule that modifies mood. Its central role in the hypoxia response and mitochondrial function and the direct impacts of SSRIs on intracellular sigma-1 receptors and the mitochondrial respiratory chain are entirely ignored in discussions about SSRI discontinuation syndrome.

However, the fact that mainstream psychiatry and pharmacology has no real answers about this, especially for those suffering protracted withdrawal lasting years, indicates that we will learn a lot more when we think outside of these traditional constraints.

Doing so leads to the conclusion that SSRI withdrawal is primarily a problem of mitochondrial dysfunction.

Unlike the one study on withdrawing from Prozac hurting the respiratory chain in mice, I do not know of any studies showing that *switching* from one SSRI to another causes mitochondrial dysfunction. But given that they have very disparate impacts on mitochondrial dysfunction, it is easy to see how this could happen, and the cases I wrote about at the beginning suggests that switching SSRIs over the course of withdrawal or after yo-yoing can play a role in the energetic dysfunction that develops.

In the next article, we will look at the ability of SSRIs themselves to cause mitochondrial dysfunction, and then we arrive at what can be done about it.

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Nattie 8h

Liked by Chris Masterjohn, PhD

Interesting. I’ve never been on SSRIs but I have been interested in their mechanisms because for about eight years now I’ve had symptoms similar to PSSD, intense anhedonia to the point of not even having a fear response, etc. I consistently got brain zaps from 5 to 10mg Adderall XR back in the day

(never took more than that in a day), and my neurological herpes prodrome features a few seconds of a very strong brain zap that feels like electric claws across the left side of my brain.

My Mitome showed iirc 11% function of Complex II + III, and low Complex IV. Based on the protocol I have been taking iron (HSV depletes it like crazy, and I have almost always had heavy periods) and L-cysteine (NAC just cements anhedonia for me) and the herpes prodrome vanished even though the iron triggered two or three outbreaks in a month. Also no sign of brain swelling, didn't dysregulate my heart rhythm like usual.

Still not sure why Adderall triggered brain zaps.

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Gail Bucker 11h

♡ Liked by Chris Masterjohn, PhD

This is very valuable information. Thank you!

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